The Hong Kong Polytechnic University
Subject Description Form

Subject Code	ME5111
Subject Title	**Interfacial Science and Engineering**
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	N.A.
Objectives	1. To provide students with knowledge of interfacial science and engineering. 2. To understand and translate the interfacial sciences into the development of advanced engineering systems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. possess state-of-the-art knowledge in the fields of interfacial science and engineering. b. apply their comprehensive knowledge of interfacial science to a wide range of engineering systems, including energy, water, environments, etc. c. extend the knowledge of interfacial science and engineering to professional practices and development in both academic and industrial communities. d. foster a good mastery of critical and creative thinking skills and generate practical and innovative solutions to novel problems. e. have recognition of the need for, and an ability to engage in life-long learning.
Subject Synopsis/ Indicative Syllabus	***Introduction–*** Definitions, concepts, and development of interfacial science and engineering. ***Classical theories –*** Basic theories. Classifications. Surface/interface effect. Surface energy. Surface tension. Wetting phenomena. Superwetting surfaces. Static/dynamic wetting. Slippery. Thermodynamics of interfaces. Capillary effect. Phase transitions. ***Nature-inspired interfacial engineering –*** Nature's surfaces/interfaces. Nature's surface principles. Scale matching principles. Gradient principles. Heterogeneous principles. Other functional surface design principles. ***Advanced interfacial techniques –*** Advanced interfacial materials. Advanced manufacturing approaches. Advanced characterization methods. Advanced visualization approaches. Advanced theoretical simulation. Machine learning. ***Multifunctional applications –*** Clean energy harvesting (Water energy, Solar energy, Thermal energy, and their hybrids for electricity generation). Energy storage (Hydrogen, Batteries). Clean water technologies (Seawater desalination, Moisture harvesting). Thermal cooling (Data center). Biological applications (Cryotherapy). Other engineering systems (Fluid transport, Adhesives, Anti-fouling, Anti-icing).

Teaching/Learning Methodology

1. The teaching and learning methods include lectures sessions, homework assignments, projects and presentation, tutorial/laboratory and examination.
2. The continuous assessment and examination are aimed at providing students with integrated knowledge required for interfacial science.
3. Technical/practical examples and problems are raised and discussed in class/tutorial sessions.
4. A team project with report and presentation will be used to enhance students' understanding of the subject contents and practice presentation skills.
5. Tutorial/Laboratory will further provide an opportunity for students to understand the importance of interfacial science in emerging water and energy technologies.

It is intended to make use of these teaching/learning methodologies to achieve the intended subject learning outcomes as indicated in the following table:

Teaching/Learning Methodology	Intended subject learning outcomes				
	a	b	c	d	e
1. Lectures	✓	✓	✓	✓	✓
2. Homework assignment	✓	✓	✓		
3. Project report and presentation				✓	✓
4. Tutorial/Laboratory				✓	

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				
		a	b	c	d	e
1. Homework assignment	15%	✓	✓	✓		
2. Project and presentation	20%				✓	✓
3. Test	15%	✓	✓	✓		
4. Examination	50%	✓	✓	✓		
Total	100 %					

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Overall Assessment:

0.50 × End of Subject Examination + 0.50 × Continuous Assessment

The continuous assessment will comprise three components: team project (20%), test (15%) and homework (15%). The team project, test and homework are aimed at evaluating their understandings on interfacial science and enhancing the integration of their knowledge learnt.
The examination (50%) will be used to assess the knowledge acquired by the students for understanding and analyzing the problems critically and independently, and to determine the degree of achieving the subject learning outcomes.

Student Study Effort Expected	Class contact:	
	▪ Lecture	36 Hrs.
	▪ Tutorial/Presentation	3 Hrs.
	Other student study effort:	
	▪ Project/Assignments	40 Hrs.
	▪ Self-study	27 Hrs.
	Total student study effort	**106 Hrs**.
Reading List and References	1. De Gennes, Pierre-Gilles, Françoise Brochard-Wyart, and David Quéré. Capillarity and wetting phenomena: drops, bubbles, pearls, waves. Springer, 2004. 2. David L Allara, Surface and Interface Science, Institute of Physics Publishing, 2024 3. Barnes and Gentle. Interfacial Science. An introduction Book, Oxford University Press, 2011. 4. Klaus Wandelt. Surface and Interface Science. Wiley-VCH Verlag GmbH & Co. KGaA, 2014 5. Harald Ibach. Physics of Surfaces and Interfaces. Springer, 2006. 6. M.C. Rastogi. Surface and Interfacial Science: Application to Engineering and Technology. Narosa Publishing House, 2003 7. Sandy B. Primrose. Biomimetics: Nature-Inspired Design and Innovation. Wiley-Blackwell, 2021. 8. Robert J. Stokes, D. Fennell Evans. Fundamentals of Interfacial Engineering, John Wiley & Sons, 1996 Journals 1. Surfaces and Interfaces 2. Droplet 3. ACS Applied Materials & Interfaces 4. Advanced Materials Interfaces 5. Progress in Surface Science	
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